

Survival Analysis of Breast Cancer Patients Using SEER Data

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1. Introduction

Breast cancer is among the most commonly diagnosed cancers worldwide and a leading cause of cancer-related death. Survival outcomes vary substantially by stage at diagnosis, patient demographics, and time period, making population-level analysis useful for understanding where disparities exist and how outcomes have changed over time.

The Surveillance, Epidemiology, and End Results (SEER) program provides individual-level registry data on cancer diagnoses, follow-up time, and vital status across a large and geographically diverse population. This analysis uses SEER data to examine overall survival among breast cancer patients diagnosed between 2010 and 2015. Specifically, it aims to:

- Describe overall survival patterns using Kaplan–Meier methods
- Compare survival across stage at diagnosis, sex, and diagnosis period
- Estimate prognostic associations with mortality using Cox proportional hazards regression

2. Data Source and Study Population

2.1 Data Source

This study uses publicly available SEER Research Data extracted using SEER*Stat. The dataset contains individual-level information on breast cancer diagnoses, follow-up time in months, and vital status at the study cutoff.

2.2 Inclusion Criteria

Cases were included if they met the following criteria:

- Primary cancer site: breast
- Malignant behavior
- Known age at diagnosis
- Year of diagnosis between 2010 and 2015
- Non-missing survival time with greater than 0 months of follow-up

Cases with unknown or zero survival time were excluded to ensure valid survival estimates.

3. Data Preparation

```
library(tidyverse)
library(survival)
```

```

library(survminer)
library(janitor)
library(cmpsk)

# Load SEER breast cancer survival data exported from SEER*Stat
seer <- read_csv("seer_breast_survival.csv", show_col_types = FALSE) %>%
  clean_names()

# Inspect structure and variable types
glimpse(seer)

## Rows: 362,254
## Columns: 6
## $ survival_months <chr> "0012", "0072", "0028", "0111~
## $ seer_cause_specific_death_classification <chr> "Alive or dead of other cause~
## $ vital_status_recode_study_cutoff_used <chr> "Dead", "Alive", "Dead", "Ali~
## $ sex <chr> "Female", "Female", "Female",~
## $ summary_stage_2000_1998_2017 <chr> "Localized", "Regional", "Unk~
## $ year_of_diagnosis <dbl> 2010, 2015, 2013, 2012, 2014,~

# Convert survival time to numeric months
seer <- seer %>%
  mutate(
    survival_months = as.numeric(survival_months),
    # Create event indicator
    # 1 = death occurred, 0 = censored (alive at cutoff)
    event = if_else(vital_status_recode_study_cutoff_used == "Dead", 1, 0),
    sex = factor(sex),
    stage = factor(summary_stage_2000_1998_2017),
    year_of_diagnosis = as.integer(year_of_diagnosis)
  ) %>%
  # Remove observations with zero or missing survival time
  filter(
    !is.na(survival_months),
    survival_months > 0
  )

# Group diagnosis years into periods to capture temporal survival trends
seer <- seer %>%
  mutate(
    dx_period = case_when(
      year_of_diagnosis <= 2011 ~ "2010-2011",
      year_of_diagnosis <= 2013 ~ "2012-2013",
      TRUE ~ "2014-2015"
    ),
    dx_period = factor(dx_period)
  )

# Check distributions across periods
table(seer$dx_period)

##
## 2010-2011 2012-2013 2014-2015
##      113899      119351      124730

```

4. Survival Analysis Methods

Overall survival was defined as the time from cancer diagnosis to death from any cause. Patients who were alive at the end of follow-up were treated as censored observations.

Survival probabilities were estimated using the Kaplan–Meier method. Differences between survival curves were evaluated using log-rank tests. Cox proportional hazards regression models were used to estimate hazard ratios (HRs) and 95% confidence intervals (CIs) for covariates of interest.

5. Kaplan–Meier Survival Analysis

5.1 Overall Survival

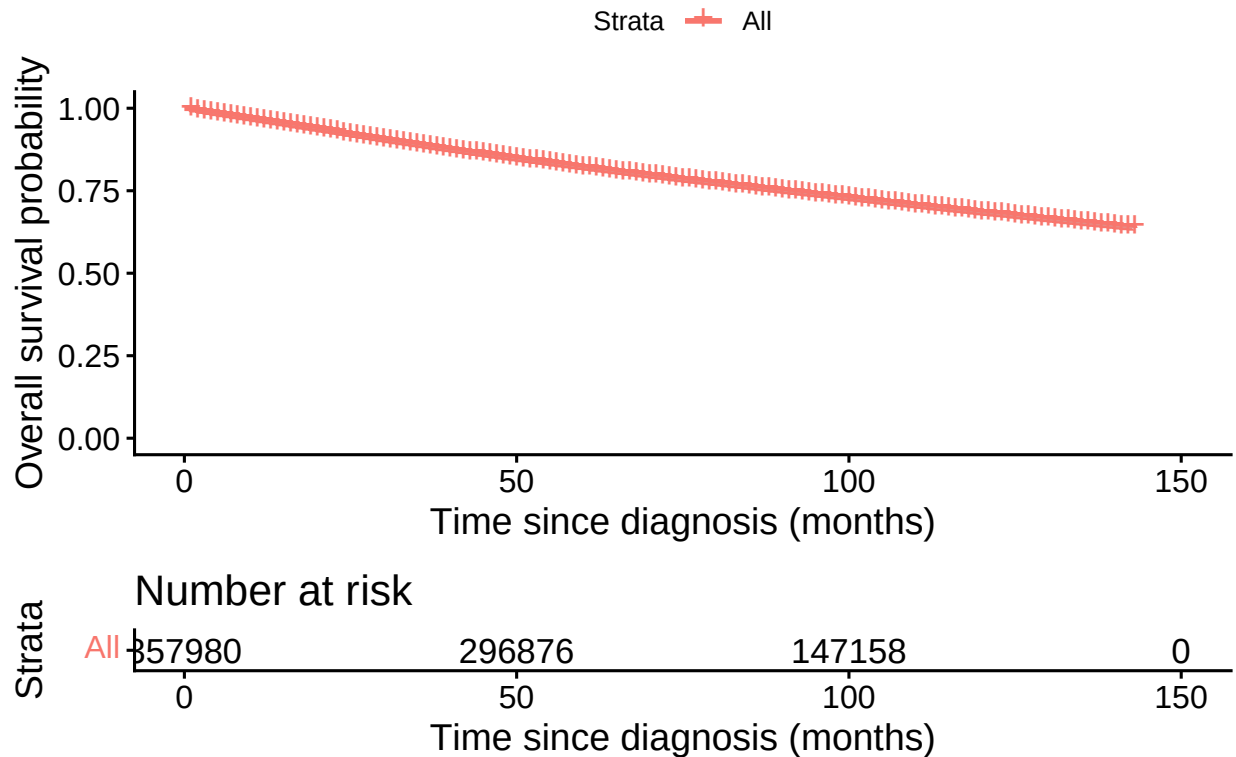
```
# Defining survival object (time-to-event framework)
surv_obj <- Surv(
  time = seer$survival_months,
  event = seer$event
)

# Fit Kaplan-Meier estimator for overall survivor
km_overall <- survfit(
  surv_obj ~ 1,
  data = seer
)

# Plot overall survival curve with confidence intervals and risk table
ggsurvplot(
  km_overall,
  data = seer,
  conf.int = TRUE,
  risk.table = TRUE,
  xlab = "Time since diagnosis (months)",
  ylab = "Overall survival probability",
  title = "Overall Survival of Breast Cancer Patients"
)

## Ignoring unknown labels:
## * fill : "Strata"
## Ignoring unknown labels:
## * fill : "Strata"
## Ignoring unknown labels:
## * fill : "Strata"
## Ignoring unknown labels:
## * fill : "Strata"
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## Ignoring unknown labels:
## * colour : "Strata"
```

Overall Survival of Breast Cancer Patients



Interpretation: The overall survival curve declines steadily from diagnosis, with the steepest drop occurring in the first 20-30 months. After that, the rate of decline slows among patients who remain in follow-up. This pattern is typical in population-based breast cancer data, where early mortality reflects a mix of aggressive disease presentations and treatment-related events.

5.2 Survival by Stage at Diagnosis

```
# Exclude unstaged cases to improve interpretability
seer_stage <- seer %>% filter(stage != "Unknown/unstaged")

# Fit stage-stratified Kaplan-Meier curves
km_stage <- survfit(
  Surv(survival_months, event) ~ stage,
  data = seer_stage
)

# Plot survival by stage (confidence interval and risk table removed for clarity)
ggsurvplot(
  km_stage,
  data = seer_stage,
  conf.int = FALSE,
  risk.table = TRUE,
  pval = TRUE,
  surv.median.line = "hv",
  censor.shape = "|",
  censor.size = 3,
```

```

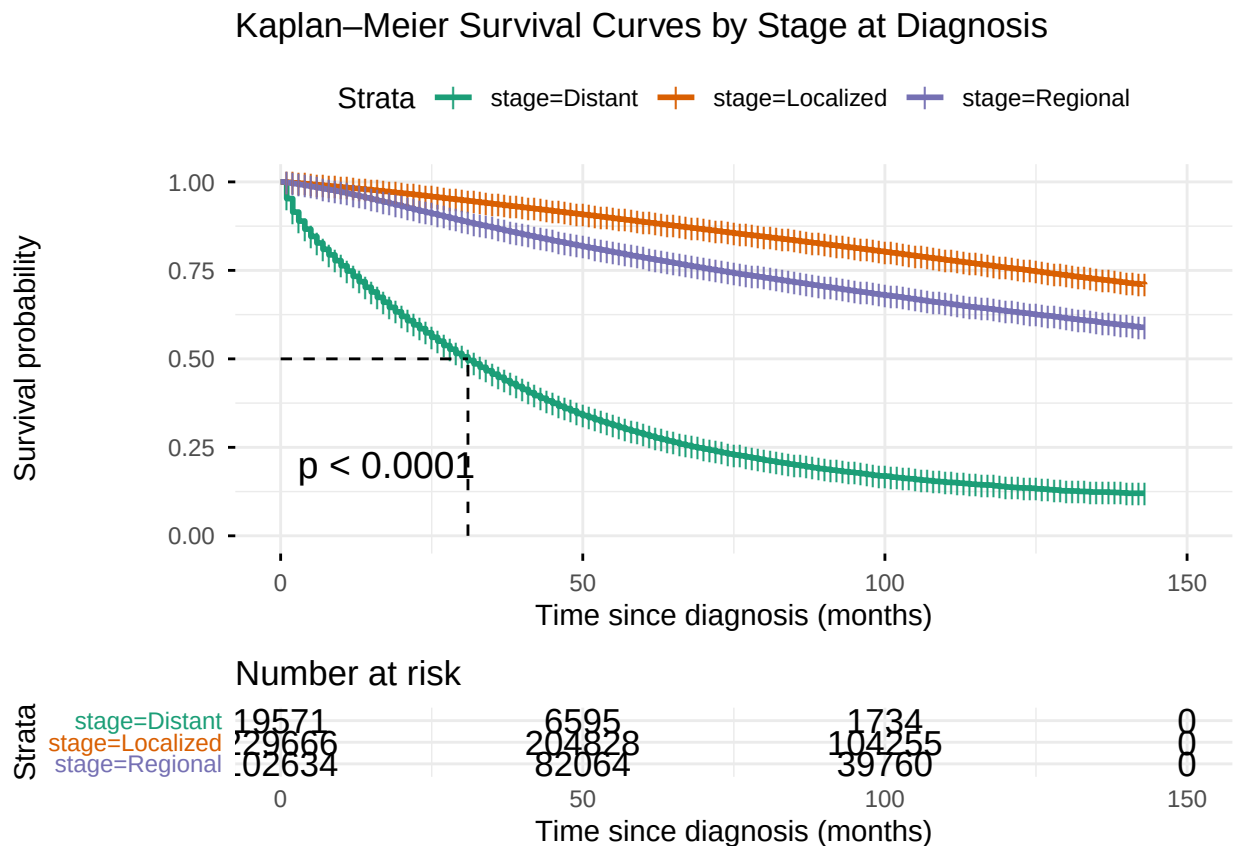
ggtheme = theme_minimal(),
palette = c("#1B9E77", "#D95F02", "#7570B3"),
xlab = "Time since diagnosis (months)",
ylab = "Survival probability",
title = "Kaplan-Meier Survival Curves by Stage at Diagnosis"
)

```

```

## Ignoring unknown labels:
## * colour : "Strata"

```



Interpretation: The stage-stratified curves separate early and stay separated, which is the expected pattern when stage is a strong prognostic factor. Localized disease shows the best survival throughout follow-up. Distant-stage patients experience rapid early mortality, with the median survival falling well below that of the other groups. Regional disease sits in between, tracking closer to localized in the early months before the gap widens over time.

The log-rank test result ($p < 0.0001$) confirms that these differences are not due to chance, though with over 350,000 observations that result is almost guaranteed regardless of effect size.

Schoenfeld residual tests flagged non-proportional hazards for stage, which is common in large registry datasets. The KM curves do not show the kind of crossing or late convergence that would indicate a serious violation. Hazard ratios from the Cox model are best read as average effects across the follow-up window rather than constant instantaneous effects.

5.3 Survival by Sex

```

# Fit Kaplan-Meier curves stratified by sex
km_sex <- survfit(

```

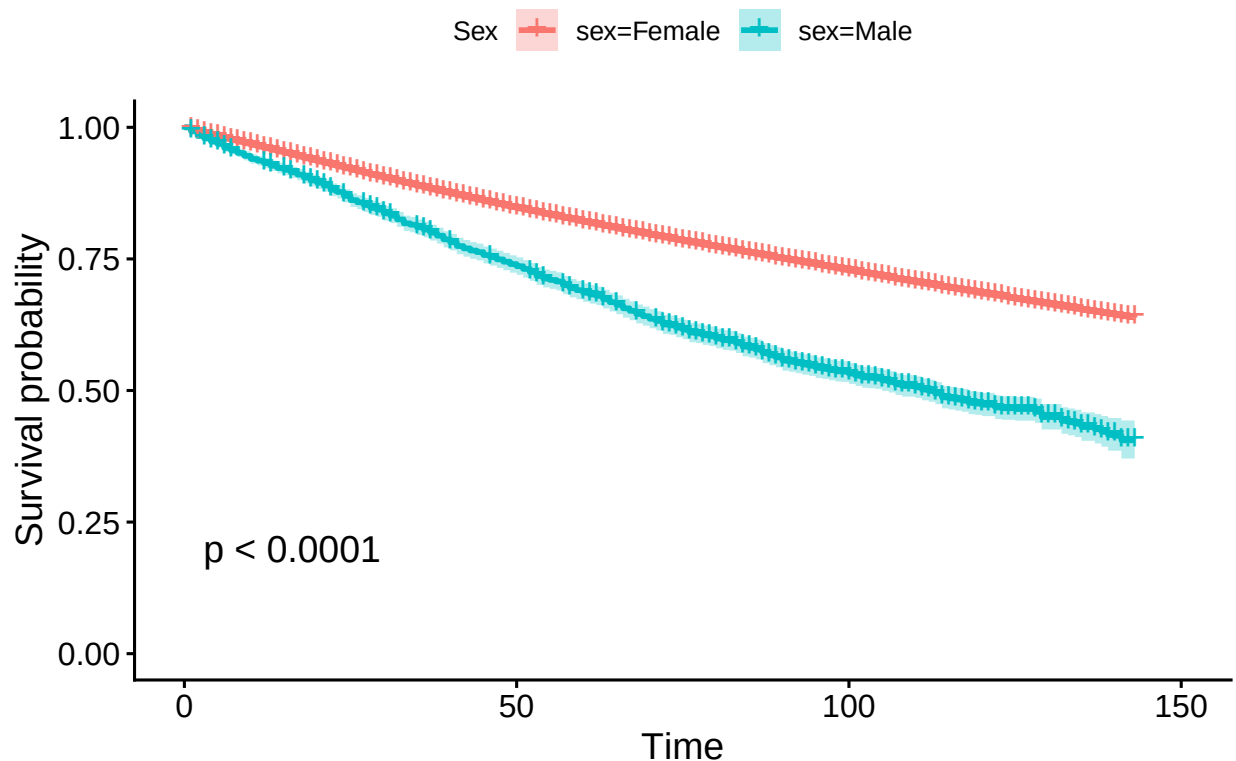
```

Surv(survival_months, event) ~ sex,
data = seer
)

# Plot survival differences by sex (Risk table removed for clarity)
ggsurvplot(
  km_sex,
  data = seer,
  conf.int = TRUE,
  risk.table = FALSE,
  pval = TRUE,
  legend.title = "Sex",
  title = "Kaplan-Meier Survival Curves by Sex"
)

```

Kaplan–Meier Survival Curves by Sex



Interpretation: Female patients show modestly better survival than male patients, but the gap is small and should be read carefully. Male breast cancer makes up a very small fraction of cases in this dataset, so the comparison is unbalanced. The difference likely reflects a mix of tumor biology, stage at diagnosis, and the fact that male patients are often diagnosed later when awareness is lower.

6. Cox Proportional Hazards Regression

6.1 Unadjusted Models

```

# Unadjusted Cox model: stage only
cox_stage <- coxph(

```

```
Surv(survival_months, event) ~ stage,
data = seer_stage
)
```

```
# Unadjusted Cox model: sex only
cox_sex <- coxph(
Surv(survival_months, event) ~ sex,
data = seer
)
```

```
summary(cox_stage)
```

```
## Call:
## coxph(formula = Surv(survival_months, event) ~ stage, data = seer_stage)
##
##      n= 351871, number of events= 97044
##
##              coef exp(coef)  se(coef)      z Pr(>|z|)
## stageLocalized    -2.214688  0.109188  0.009265 -239.0  <2e-16 ***
## stageRegional     -1.666763  0.188857  0.009712 -171.6  <2e-16 ***
## stageUnknown/unstaged      NA          NA  0.000000    NA      NA
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##              exp(coef) exp(-coef) lower .95 upper .95
## stageLocalized      0.1092      9.159    0.1072    0.1112
## stageRegional       0.1889      5.295    0.1853    0.1925
## stageUnknown/unstaged      NA          NA          NA          NA
##
## Concordance= 0.634 (se = 0.001 )
## Likelihood ratio test= 40010 on 2 df,  p=<2e-16
## Wald test              = 57134 on 2 df,  p=<2e-16
## Score (logrank) test = 78565 on 2 df,  p=<2e-16
```

```
summary(cox_sex)
```

```
## Call:
## coxph(formula = Surv(survival_months, event) ~ sex, data = seer)
##
##      n= 357980, number of events= 100587
##
##              coef exp(coef) se(coef)      z Pr(>|z|)
## sexMale 0.6693    1.9528    0.0279 23.99  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##              exp(coef) exp(-coef) lower .95 upper .95
## sexMale      1.953      0.5121    1.849    2.063
##
## Concordance= 0.503 (se = 0 )
## Likelihood ratio test= 468.2 on 1 df,  p=<2e-16
## Wald test              = 575.6 on 1 df,  p=<2e-16
## Score (logrank) test = 597.4 on 1 df,  p=<2e-16
```

Interpretation: Unadjusted models indicate that advanced stage at diagnosis and male sex are associated

with increased hazards of death. Localized stage was used as the reference category.

6.2 Multivariable Model

```
# Multivariable Cox model adjusting for stage, sex and diagnosis period
cox_multi <- coxph(
Surv(survival_months, event) ~ stage + sex + dx_period,
data = seer_stage
)

summary(cox_multi)
```

```
## Call:
## coxph(formula = Surv(survival_months, event) ~ stage + sex +
##       dx_period, data = seer_stage)
##
##      n= 351871, number of events= 97044
##
##              coef exp(coef)  se(coef)      z Pr(>|z|)
## stageLocalized    -2.212343  0.109444  0.009269 -238.685 < 2e-16 ***
## stageRegional     -1.668890  0.188456  0.009714 -171.800 < 2e-16 ***
## stageUnknown/unstaged      NA         NA  0.000000      NA      NA
## sexMale            0.550067  1.733368  0.028347  19.405 < 2e-16 ***
## dx_period2012-2013 -0.015193  0.984921  0.007786  -1.951  0.051 .
## dx_period2014-2015 -0.053122  0.948264  0.008406  -6.320 2.62e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##              exp(coef) exp(-coef) lower .95 upper .95
## stageLocalized      0.1094      9.1371      0.1075      0.1115
## stageRegional       0.1885      5.3063      0.1849      0.1921
## stageUnknown/unstaged      NA         NA         NA         NA
## sexMale             1.7334      0.5769      1.6397      1.8324
## dx_period2012-2013   0.9849      1.0153      0.9700      1.0001
## dx_period2014-2015   0.9483      1.0546      0.9328      0.9640
##
## Concordance= 0.637 (se = 0.001 )
## Likelihood ratio test= 40369 on 5 df,  p=<2e-16
## Wald test              = 57586 on 5 df,  p=<2e-16
## Score (logrank) test = 79041 on 5 df,  p=<2e-16
```

Interpretation: Stage dominates the model. Compared to distant-stage disease (the reference), localized patients have roughly one-tenth the hazard of death (HR ~0.11), and regional patients have about one-fifth (HR ~0.19). These are large effects that hold up after adjusting for sex and diagnosis period.

Sex contributes meaningfully: male patients have about 1.73 times the hazard of female patients after adjustment. The diagnosis period effect is smaller but present – patients diagnosed in 2014-2015 have modestly lower hazards than those diagnosed in 2010-2011, suggesting some improvement in outcomes over this window, though the magnitude is small (HR ~0.95).

6.3 Proportional Hazards Assumption

```
# Test proportional hazards assumption using Schoenfeld residuals
ph_test <- cox.zph(cox_multi)
```



```
ph_test
```

```
##           chisq df      p
## stage      2249.899  2 < 2e-16
## sex         11.204  1 0.00082
## dx_period    0.533  2 0.76613
## GLOBAL      2270.096  5 < 2e-16
```

Interpretation: The Schoenfeld residual tests flag stage and sex as non-proportional (both $p < 0.001$), but with $n > 350,000$ this test will reject proportionality for almost any real deviation, including trivial ones. The KM curves for stage do not cross and do not converge late in follow-up, which is what a meaningful violation would look like. The Cox HRs are interpreted as time-averaged estimates rather than constant instantaneous effects.

6.4 Competing Risks: Breast Cancer vs. Other-Cause Mortality

```
library(cmprsk)

# Build cause-specific event indicator using SEER COD variable
# 1 = breast cancer death (event of interest)
# 2 = other-cause death (competing event)
# 0 = censored (alive at cutoff)

seer_cr <- seer_stage %>%
  mutate(
    event_cr = case_when(
      seer_cause_specific_death_classification ==
        "Dead (attributable to this cancer dx)" ~ 1,
      vital_status_recode_study_cutoff_used == "Alive" ~ 0,
      TRUE ~ 2
    )
  )

table(seer_cr$event_cr)

##
##      0      1      2
## 254827 48662 48382

# Cumulative incidence by stage
# Gray's test is computed automatically by cuminc()

ci_stage <- cuminc(
  ftime = seer_cr$survival_months,
  fstatus = seer_cr$event_cr,
  group = seer_cr$stage
)

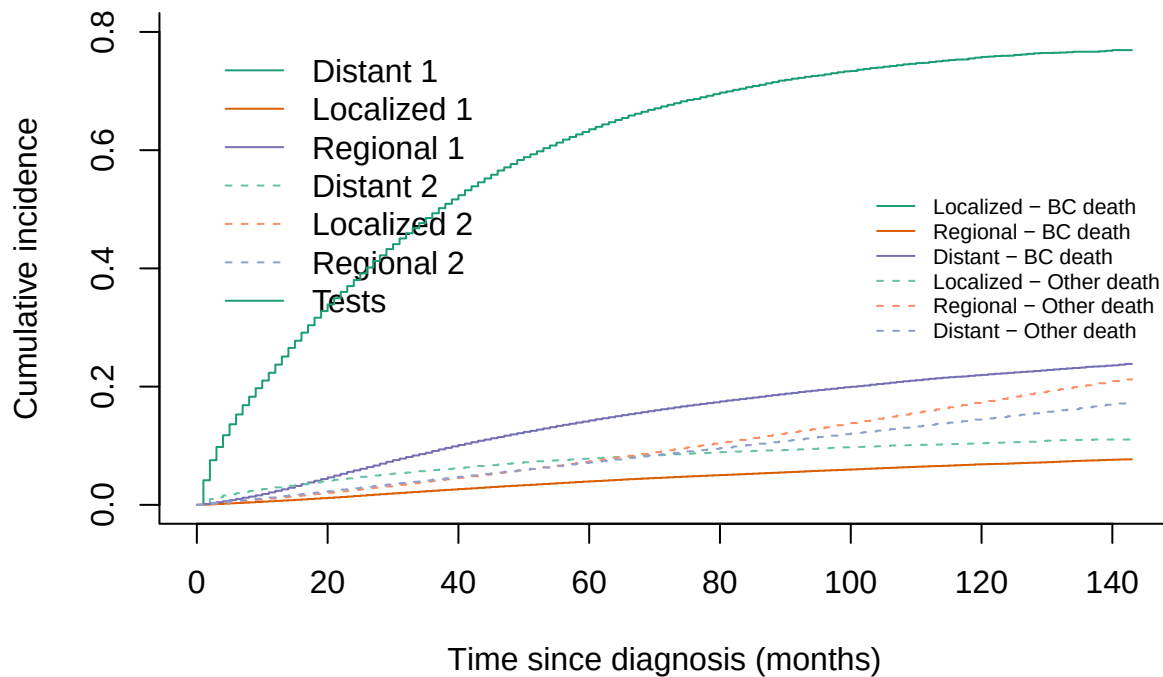
# Plot CIF for breast cancer death (event 1) by stage only
# Pull just the event=1 curves from the cuminc object
plot(
  ci_stage,
  curvlab = names(ci_stage),
  col = c("#1B9E77", "#D95F02", "#7570B3",
```

```

      "#66C2A5", "#FC8D62", "#8DA0CB"),
lty    = c(1, 1, 1, 2, 2, 2),
xlab    = "Time since diagnosis (months)",
ylab    = "Cumulative incidence",
main    = "Cumulative Incidence by Stage: Breast Cancer vs. Other-Cause Death",
ylim    = c(0, 0.8)
)
legend("right",
      legend = c("Localized - BC death", "Regional - BC death",
                  "Distant - BC death",
                  "Localized - Other death", "Regional - Other death",
                  "Distant - Other death"),
      col = c("#1B9E77", "#D95F02", "#7570B3",
              "#66C2A5", "#FC8D62", "#8DA0CB"),
      lty = c(1, 1, 1, 2, 2, 2),
      cex = 0.65,
      bty = "n")

```

Cumulative Incidence by Stage: Breast Cancer vs. Other-Cause Dea



```

# Gray's test results -- tests whether CIFs differ across stage groups
# for breast cancer death specifically
print(ci_stage$Tests)

```

```

##          stat pv df
## 1 81656.8236  0  2
## 2  479.0327  0  2

```

Interpretation:

The CIF plot separates two things the KM analysis could not: dying from breast cancer versus dying from something else. For distant-stage patients, breast cancer death accumulates rapidly, reaching roughly 0.75 by the end of follow-up. Localized patients show a very different picture, with breast cancer-specific cumulative incidence staying below 0.10 throughout. Regional disease sits between these two.

Other-cause mortality (dashed lines) rises gradually across all stage groups and follows a similar trajectory regardless of stage, which makes sense since background mortality risk does not depend on cancer stage. For localized patients, other-cause death is actually more common than breast cancer death over a long enough horizon, which is something the standard KM analysis obscures.

Gray’s test confirms that the CIFs differ significantly across stage groups for breast cancer death (stat = 81,657, $p < 0.001$) and other-cause death (stat = 479, $p < 0.001$). The key methodological point: KM treats other-cause deaths as censored observations, which inflates the estimated probability of breast cancer death. The CIF does not make that assumption and gives a more accurate read of absolute risk.

7. Limitations

- This study used SEER public-use registry data, which do not contain detailed information on treatment modalities (e.g., chemotherapy regimens, targeted therapies, radiation dose) or comorbid conditions. As a result, the observed associations between stage, sex, and survival may be partially confounded by unmeasured clinical and treatment-related factors.
- The primary Cox regression models used overall survival as the endpoint, meaning deaths from causes unrelated to breast cancer were included as events. To address this, a competing risks analysis was conducted to separately estimate breast cancer-specific and other-cause mortality using the SEER cause-specific death classification variable. Cumulative incidence functions and Gray’s test were used in place of Kaplan-Meier estimates and log-rank tests, which are known to overestimate cause-specific risk in the presence of competing events.
- Information on age was limited to recoded categories in the public-use dataset, preventing modeling of age as a continuous covariate. This limitation may reduce precision in estimating age-related survival differences and restrict more flexible modeling of age effects.
- Cases with incomplete follow-up or zero survival time were excluded to ensure valid time-at-risk definitions. While methodologically appropriate, this exclusion may lead to a slight underestimation of very early mortality immediately following diagnosis.
- The Cox proportional hazards model assumes that hazard ratios remain constant over time. Although formal tests and diagnostic plots did not indicate major violations of this assumption, minor departures from proportionality cannot be fully ruled out, particularly in long-term follow-up. As such, hazard ratio estimates should be interpreted as average effects over the study period.

8. Implications

Stage at diagnosis is the clearest driver of breast cancer survival in this dataset, which points toward early detection as the most direct lever for improving population-level outcomes. Shifts toward earlier-stage diagnosis at the population level would likely do more for survival rates than incremental improvements in treatment for advanced disease.

The sex difference is smaller and harder to interpret cleanly. Male breast cancer is rare, the comparison group is small, and the gap in survival likely reflects multiple overlapping factors including later diagnosis, different tumor biology, and lower clinical awareness. It is worth tracking but should not be over-interpreted from registry data alone.

The modest improvement in survival for more recently diagnosed patients is real but small. Whether it reflects better screening, improved systemic therapies, or both cannot be determined from these data alone. Answering that question would require treatment variables that SEER public-use data does not include.

9. Conclusion

This analysis used SEER registry data to examine breast cancer survival patterns across stage, sex, and diagnosis period. Stage at diagnosis was by far the strongest predictor of outcomes. Localized disease carried substantially better survival than regional or distant disease, a difference that appeared early and persisted across the full follow-up window. The multivariable Cox model confirmed this after adjustment, with distant-stage patients facing roughly nine times the mortality hazard of localized patients.

The competing risks analysis added precision to the overall survival findings. For localized-stage patients, other-cause mortality is a meaningful competing event, and the standard KM approach would overestimate breast cancer-specific risk in that group. The cumulative incidence functions give a more honest picture of what patients are actually dying from.

The main limitation is the absence of treatment data. Stage associations in this analysis could partly reflect differences in treatment received rather than disease biology alone. Future work with restricted-access SEER data or linked claims data could address this.